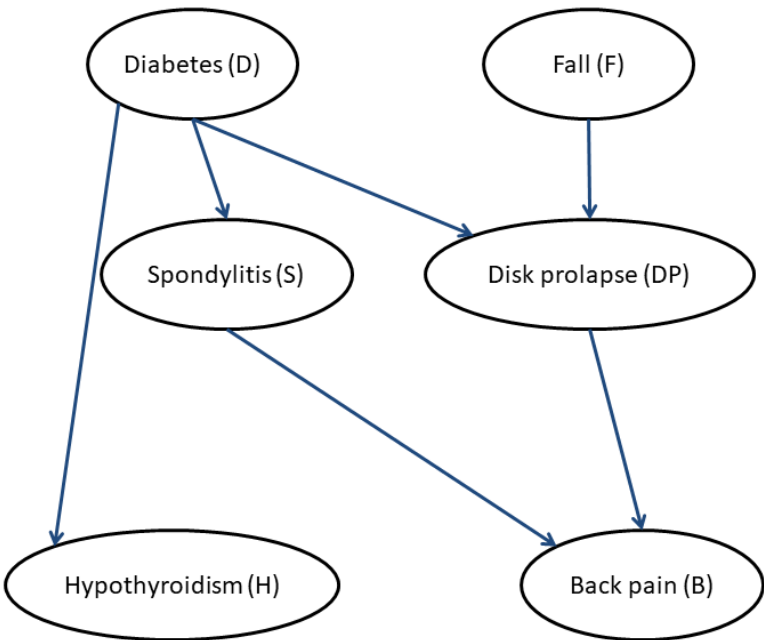


July 2021 Term
CSE 471 – Machine Learning
Class Test# 2 Time: 20 minutes

Name: _____ Student No. _____

No extra sheet shall be provided.

1. Consider the Bayesian Network (BN) structure along with the various survey results as specified below.



- 1. In a survey, 1.0% respondents selected “YES” to having encountered a significant fall at any point during their life.
- 2. The global diabetes prevalence in 2019 is estimated to be 9.3%
- 3. Table 1 summarizes another survey results on Diabetic and Non-diabetic people. The number of data points are mentioned directly in a cell, or inside parenthesis, as appropriate. As an example, from Table 1, we can see that 30 out of 100 diabetic patients have hypothyroidism. Of 50 patients having diabetes as well as a fall event in their life, only 20 had disk prolapse.
- 4. Table 2 gives the CPT of back pain event.

Table 1: Results of a survey				
	Hypothyroidism	Spondylitis	Disk prolapse	
Diabetic (100)	30	25	20	Fall (50)
			12	No fall (50)
Non-diabetic (100)	10	15	10	Fall (50)
			4	No fall (50)

Table 2: CPT of Back pain		
S	DP	P(B=true S, DP)
t	t	0.90
t	f	0.65
f	t	0.75
f	f	0.30

Based on the above information

- 1. Construct the CPT table for each node of the network. (10)
- 2. Calculate the posterior probability of a person having diabetes, given s/he never had a fall, has back pain but no hypothyroidism. (10)

Answer

- 1. $P(D=\text{true}) = 0.093$
- 2. $P(F=\text{true}) = 0.01$

D	P(S=true D)	D	P(H=true D)	D	F	P(DP=true D, F)
t	0.25	t	0.30	t	t	0.40
f	0.15	f	0.10	t	f	0.24
				f	t	0.20
				f	f	0.08

$$\begin{aligned}
P(d|b, \neg h, \neg f) &= \alpha P(d, b, \neg h, \neg f) \\
&= \alpha \sum_{dp} \sum_s P(d, b, \neg h, s, dp, \neg f) \\
&= \alpha \sum_{dp} \sum_s P(d)P(\neg f)P(s|d)P(\neg h|d)P(dp|d, \neg f)P(b|s, dp) \\
&= \alpha P(d)P(\neg f)P(\neg h|d) \sum_s P(s|d) \sum_{dp} P(dp|d, \neg f)P(b|s, dp) \\
&= \alpha \times 0.093 \times 0.99 \times 0.70 \\
&\quad \times [0.25 \times [0.24 \times 0.9 + 0.76 \times 0.65] + 0.75 \times [0.24 \times 0.75 + 0.76 \times 0.30]] \\
&= \alpha 0.064449 \times [0.25 \times 0.71 + 0.75 \times 0.408] = 0.0311610915\alpha
\end{aligned}$$

$$\begin{aligned}
P(\neg d|b, \neg h, \neg f) &= \alpha P(\neg d, b, \neg h, \neg f) \\
&= \alpha \sum_{dp} \sum_s P(\neg d, b, \neg h, s, dp, \neg f) \\
&= \alpha \sum_{dp} \sum_s P(\neg d)P(\neg f)P(s|\neg d)P(\neg h|\neg d)P(dp|\neg d, f)P(b|s, dp) \\
&= \alpha P(\neg d)P(\neg f)P(\neg h|\neg d) \sum_s P(s|\neg d) \sum_{dp} P(dp|\neg d, f)P(b|s, dp) \\
&= \alpha \times 0.907 \times 0.99 \times 0.90 \\
&\quad \times [0.15 \times [0.08 \times 0.9 + 0.92 \times 0.65] + 0.85 \times [0.08 \times 0.75 + 0.92 \times 0.30]] \\
&= \alpha 0.808137 \times [0.15 \times 0.67 + 0.85 \times 0.336] = 0.3120216957\alpha
\end{aligned}$$

Therefore, $0.0311610915\alpha + 0.3120216957\alpha = 1 \rightarrow \alpha = 2.9138990569979262642925466630163$

Therefore, $P(d|b, \neg h, \neg f) = 0.0311610915 \times 2.9138991 = 0.0908$